

HackerRank Algorithms & GitHub Coding Portfolio

Student Details

Name	Nikhi Rathod
Student ID / USN	R25EF167
Semester	3rd Semester
Programming Language	C++14

Profile & Repository Links

HackerRank Profile	https://www.hackerrank.com/profile/nikhimaha13
GitHub Repository	https://github.com/nikhi-sys/HackerRank-3rdSem-Algorithm-Portfolio

Completed Challenges

No.	Problem	Topic	Status
1	Mini-Max Sum	Arrays / Implementation	Accepted
2	Birthday Cake Candles	Arrays / Counting	Accepted
3	Insertion Sort - Part 1	Sorting	Accepted
4	Binary Search (Intro to Tutorial Challenges)	Searching	Accepted
5	Mark and Toys	Greedy / Sorting	Accepted

HackerRank Evidence

Accepted-submission evidence (screenshots 1–5).

Screenshot 1 — Mini-Max Sum



Screenshot 4 — Birthday Cake Candles — Accepted



Screenshot 2 — Mini-Max Sum — Accepted



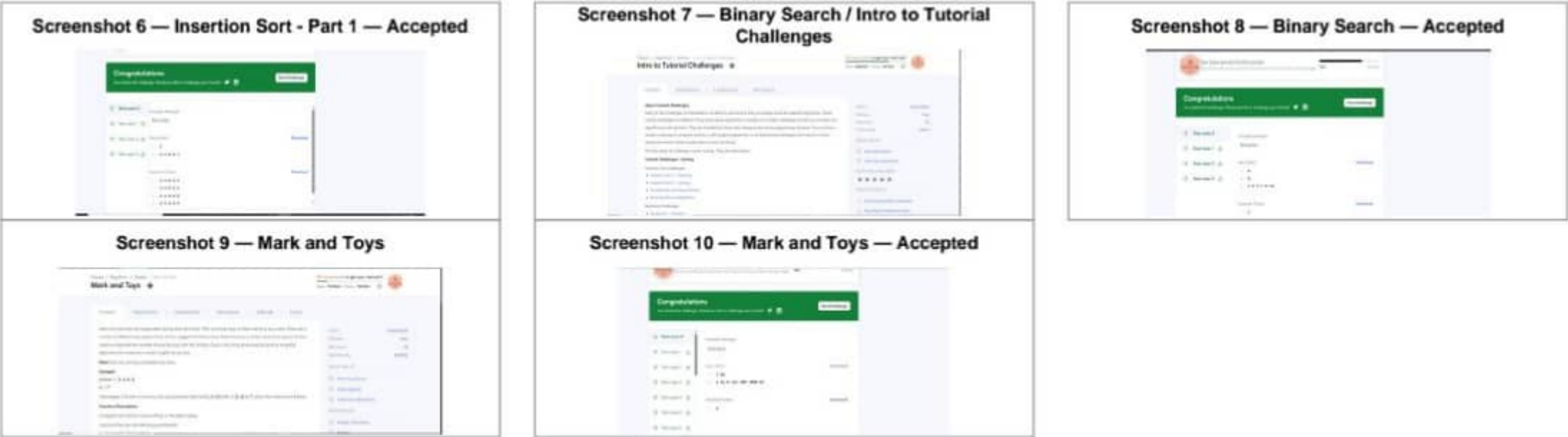
Screenshot 5 — Insertion Sort - Part 1



Screenshot 3 — Birthday Cake Candles



HackerRank Evidence — Continued



Algorithm Analysis & Reflection

Time & Space Complexity Summary

Problem	Approach	Time	Auxiliary Space
Mini-Max Sum	Single traversal; track total, min and max	O(N)	O(N)
Birthday Cake Candles	Single traversal; track maximum and count	O(N)	O(N)
Insertion Sort - Part 1	Shift larger elements and insert final value	O(N)	O(N)
Binary Search	Repeatedly halve sorted search range	O(log N)	O(1)
Mark and Toys	Sort prices; greedily buy cheapest toys	O(N log N)	O(log N)*

* Mark and Toys: O(log N) refers to typical auxiliary stack space associated with C++ sorting; the input vector itself uses O(N) storage.

Algorithmic Techniques Learned

Arrays / Implementation	Efficient traversal and tracking of required values.
Counting	Maintaining a maximum and its frequency in one pass.
Insertion / Sorting	Shifting larger elements to insert a value correctly.
Binary Search	Reducing the search interval by half at each step.
Greedy + Sorting	Sorting by price and selecting the cheapest feasible items.

Learning Reflection

Through this activity, I developed a better understanding of how algorithmic techniques can be selected according to the structure of a problem. The Mini-Max Sum and Birthday Cake Candles problems helped me practice array traversal, tracking values, and counting occurrences efficiently. Insertion Sort - Part 1 helped me understand how elements are shifted to place a value in its correct position. Binary Search introduced the importance of using a sorted array to reduce the search space by half at every step, giving a logarithmic time complexity. Mark and Toys demonstrated the greedy approach, where sorting prices and selecting the cheapest available items helps maximize the number of purchases within a fixed budget. I also learned to analyze both time and auxiliary space complexity instead of considering only whether a program produces the correct output. Uploading the solutions to GitHub helped me organize my work into separate problem folders and maintain a clear coding portfolio.

Verification & Badge Evidence

HackerRank profile: <https://www.hackerrank.com/profile/nikhimaha13>
GitHub repository: <https://github.com/nikhi-sys/HackerRank-3rdSem-Algorithm-Portfolio>
Badge evidence: Problem Solving badge shown in the uploaded evidence.

